Mapping Potential Salt Deposition and Exposure Risk for *Drosera capillaris* and *Drosera rotundifolia* near the Twin Pines Mine, Georgia.

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INTRODUCTION

- Okefenokee Swamp is a critical element of Southeast Georgia because it acts as a hotspot for biodiversity.
- The proposed mining from Twin Pines was right along the edge of Okefenokee (Figure 2).
- Twin Pines mining proposed evaporating salts into the atmosphere, which could impact plants in the swamp.
- Sundews (*Drosera capillris* (Figure 1) and *Drosera rotundifolia)* were chosen for this research project because they are extremely sensitive plants making them serve as bioindicators for wetlands.
- Wetland health is essential for Southeast Georgia because they prevent floods and act as a carbon sink.

Figure 1: Image of Pink Sundew *(Drosera capillaris)*

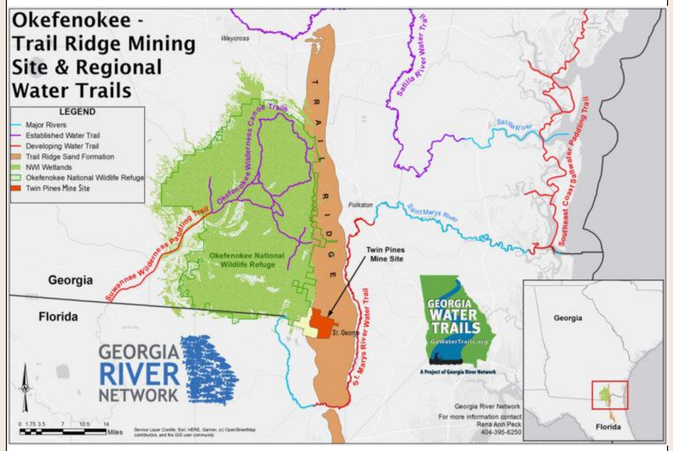

Figure 2: Site Map of Proposed Mining

OBJECTIVES

- Estimate potential salt deposition patterns around the Twin Pines site.
- Assess the exposure risk to nearby *Drosera capillaris* and *Drosera rotundifolia* habitat.

METHODS

- ArcGIS Pro was used to analyze the salt deposition patterns from the Twin Pines proposed site and examine the risk to *Drosera capillaris* and *Drosera rotundifolia*
- Data on Twin Pines mining proposal was gathered from a published article (source)
- Salt exposure data was gathered from NOAA Hysplit Diversion Footprint to show how far the salt exposure risk would be
- Potential *Drosera sp.* habitat was gathered from published online sources of ideal habitats and referencing them to areas with similar environmental conditions within the range of exposure

RESULTS

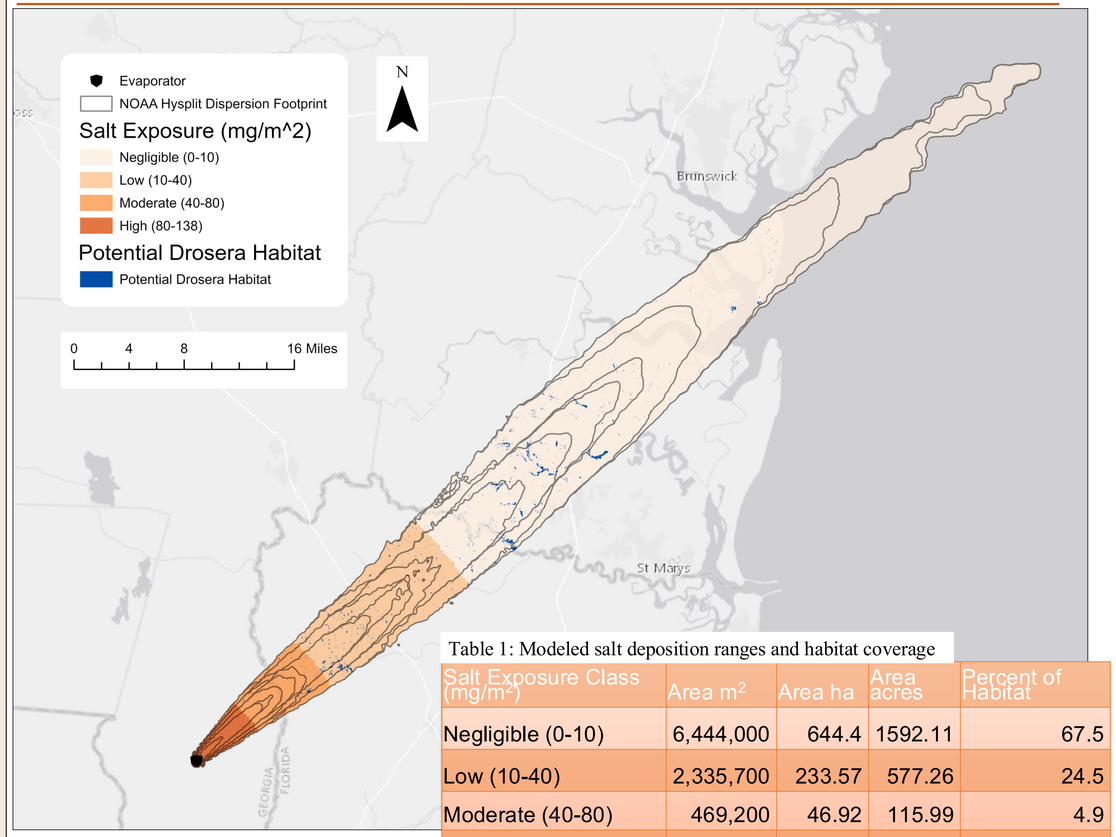

Figure 3: Salt exposure in Trail Ridge from where evaporator would be installed.

Table 1: Modeled salt deposition ranges and habitat coverage

Salt Exposure Class (mg/m²)	Area m²	Area ha	Area acres	Percent of Habitat
Negligible (0-10)	6,444,000	644.4	1592.11	67.5
Low (10-40)	2,335,700	233.57	577.26	24.5
Moderate (40-80)	469,200	46.92	115.99	4.9
High (80-138)	303,500	30.35	74.99	3.2
Total	9,552,400	955.24	2360.35	100

CONCLUSION

- The NOAA dispersion model found a linear northeast-directed plume within a 12-hour modeling cycle and the high exposure class extends 8 km form the proposed mine surpassing our hypothesis of 5 km (Figure 3).
- Based on the Okefenokee peat structure and assumed daily deposition, salt mixing was determined to be in the top 0.1 m layer. Below this zone, the diffusion dominates and vertical mixing is minimal (Cohen et al., 1973).
- Non-halophyte wetland plants such as *Drosera spp.* can show stress at 1 ppt or ~100 mg/m2 of salt exposure shown in Table 1. (Hai-shun et al., 2020; Moreira et al., 2023).
- The high exposure class are likely to experience decrease in photosynthetic pigments, increased oxidative stress, and altered nutrient uptake.
- 1 ppt salt exposure may also reduce *Drosera* abundance and seedling richness (Pereira et al., 2018). Consequently, shifts in insect community structure are possible.

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REFERENCES

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